📘 Societal Impacts

1. Digital Footprint

What It Is

Your digital footprint is the trail of data you leave behind whenever you use the internet. Every search, social media post, online purchase, website visit, and app interaction contributes to this footprint. It includes both information you deliberately share and data collected about you without your direct awareness.

Types of Digital Footprints

Type	Description	Examples
Active Footprint	Data you intentionally share online	Social media posts, comments, online forms, reviews
Passive Footprint	Data collected automatically without your direct action	Cookies, IP address logs, browsing history, geolocation data

Why It Matters

Your digital footprint can be used by:

- **Employers** — screening job candidates
- **Educational institutions** — admissions decisions
- **Advertisers** — targeted marketing
- **Cybercriminals** — identity theft and phishing attacks
- **Government agencies** — surveillance and law enforcement

Managing Your Digital Footprint

1. **Use strong passwords** and enable two-factor authentication to prevent unauthorized access
2. **Adjust privacy settings** on social networks to control who sees your information
3. **Regularly clear browser history and cookies** to reduce tracking
4. **Be cautious about sharing personal data** — once posted, it can be difficult to remove

5. **Delete inactive accounts** that could be exploited by hackers
6. **Use a VPN** to enhance connection anonymity

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2. Net and Communication Etiquettes (Netiquette)

Definition

Netiquette combines “net” (Internet) and “etiquette” — the rules of correct and polite behavior among people using the Internet.

Core Principles

> ☀️ **The Golden Rule Online:** Never say anything online that you would not say face-to-face. Remember that online interactions are almost always recorded and can resurface years later.

Platform Awareness: Consider the communication dynamics of each platform. What you share on WhatsApp may not be appropriate for email or a professional forum.

Key Netiquette Rules

Rule	Explanation
Respect others’ privacy	Do not share personal information or photographs of third parties without permission
Be cordial	Greet at the start, say goodbye at the end, avoid sarcasm and insults
Be brief	Limit messages to what is necessary — you are using the recipient’s time
Check before sharing	Share only verified information from reliable sources
Avoid cyberbullying	Do not spread rumors, lies, or hurtful jokes; report such behavior
Proofread	Check grammar and spelling; avoid ALL CAPS (equivalent to shouting)
Mind file sizes	Large attachments consume server storage and recipient bandwidth

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3. Data Protection

What Is Data Protection?

Data protection refers to the legal and technical measures designed to safeguard personal data from unauthorized access, misuse, disclosure, or destruction.

India's Digital Personal Data Protection Act, 2023 (DPDPA)

India enacted the DPDPA to establish a comprehensive legal framework for processing digital personal data.

Key Provisions:

- **Consent Requirement:** Personal data can only be processed for lawful purposes with the consent of the individual (Data Principal) or for certain legitimate uses
- **Right to Information:** Individuals have the right to obtain a summary of their personal data and processing activities from the Data Fiduciary
- **Security Obligations:** Data Fiduciaries must take reasonable security safeguards to prevent data breaches
- **Breach Notification:** In case of a breach, the Data Fiduciary must inform the Data Protection Board and each affected individual
- **Penalties:** Monetary penalties can reach up to **₹250 crore** for serious violations

Special Protections for Children:

- Verifiable parental consent required before processing children's data
- Prohibition on tracking, behavioral monitoring, and targeted advertising directed at children

Practical Data Protection Measures

- Use encryption for sensitive communications
- Enable two-factor authentication on all accounts
- Regularly update software to patch security vulnerabilities
- Be cautious with public Wi-Fi networks
- Review app permissions before installation

4. Intellectual Property Rights (IPR)

Definition

Intellectual Property Rights are legal protections granted to creators and inventors for their creations of the mind — inventions, literary and artistic works, designs, symbols, names, and images used in commerce.

Types of IPR

Type	Protects	Duration	Registration
Copyright	Literary, artistic, musical, dramatic works; software	Life of author + 50-70 years	Automatic upon creation
Patents	Inventions (products or processes)	20 years	Required
Trademarks	Signs, logos, brand names distinguishing goods/services	10 years, renewable indefinitely	Generally required
Industrial Designs	Aesthetic/ornamental aspects of products	10-15 years	Required
Geographical Indications	Products with specific geographical origin	Indefinite (until cancelled)	Required
Trade Secrets	Confidential business information	Indefinite (if kept secret)	No registration; must maintain secrecy

Copyright: Key Concepts

Copyright grants the holder **exclusive rights** to:

- Reproduce the work
- Prepare derivative works (modifications)
- Distribute copies to the public

Important: Copyright applies automatically as soon as a work is created and saved. No registration is needed, though copyright notices are recommended.

5. Plagiarism

Definition

Plagiarism is presenting someone else's work, ideas, or words as your own without proper attribution. It is an **ethical violation** and may also constitute **copyright infringement** if the work is protected.

Types of Plagiarism

- **Direct plagiarism:** Copying word-for-word without quotation marks or citation
- **Mosaic plagiarism:** Mixing copied phrases with your own words without proper citation
- **Self-plagiarism:** Reusing your own previously submitted work without disclosure
- **Accidental plagiarism:** Unintentionally failing to cite sources properly

Plagiarism vs. Copyright Infringement

Aspect	Plagiarism	Copyright Infringement
Nature	Ethical violation	Legal violation
Concern	Attribution and honesty	Economic rights of creator
Enforcement	Academic/professional consequences	Legal action by rights holder

You can plagiarize **public domain** works (no copyright) if you fail to cite them, but you cannot infringe copyright on public domain works.

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6. Licensing and Copyright

Copyright Licensing

A license is permission granted by the copyright holder allowing others to exercise rights (copying, modification, distribution) that would otherwise be exclusive to the holder.

Key Principles:

- The copyright holder decides who can use the work and under what terms
- Using a work outside the license terms constitutes **infringement**

- Licenses can be tailored — the holder may grant exceptions for specific uses

Common License Types

License Type	Description
Proprietary	All rights reserved; use restricted by the vendor
Free/Open Source	Source code available; users can modify and redistribute
Creative Commons	Standardized public licenses for creative works

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7. Free and Open Source Software (FOSS)

Definition

FOSS refers to software that grants users the freedom to run, study, modify, and distribute the software and its source code. "Free" refers to **freedom, not price** — software can be sold and still be free software.

The Four Freedoms

1. **Freedom to run** the program for any purpose
2. **Freedom to study** how the program works and modify it
3. **Freedom to redistribute** copies
4. **Freedom to distribute** modified versions to others

FOSS vs. Proprietary Software

Aspect	FOSS	Proprietary
Source code	Available to all	Hidden
Modification	Permitted	Prohibited
Distribution	Permitted	Restricted
Cost	Often free, but not necessarily	Usually paid
Examples	Linux, LibreOffice, VLC, Firefox	Windows, Microsoft Office, Adobe Photoshop

The Right to Fork

If maintainers take a project in a bad direction, the community can copy the code and continue on their own path. LibreOffice and OpenTofu exist because of this right.

Key Organizations

- **Free Software Foundation (FSF):** Founded by Richard Stallman; maintains the GNU General Public License (GPL)
- **Open Source Initiative (OSI):** Founded in 1998 to make free software more commercially attractive

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8. Cybercrime and Cyber Laws

Definition

Cybercrime refers to criminal activities that involve computers, networks, or digital devices as either the **tool** or **target** of the crime.

Categories of Cybercrime

By Target:

Category	Description	Examples
Individual	Crimes against persons	Cyberbullying, identity theft, online harassment
Property	Crimes involving financial gain	Phishing, credit card fraud, ransomware
Government	Crimes against state	Hacking government systems, cyberterrorism

By Nature:

Type	Description
Cyber-dependent	Crimes that can only be committed using technology (e.g., hacking)
Cyber-enabled	Traditional crimes scaled up by technology (e.g., fraud, stalking)

9. Hacking

Definition

Hacking is the **unauthorized access** to computers or networks by exploiting security vulnerabilities or bypassing security measures. Hackers may steal data, disrupt operations, or cause damage.

Types of Hackers

- **White Hat:** Ethical hackers who test security with permission
- **Black Hat:** Malicious hackers who break in for personal gain
- **Grey Hat:** May violate laws without malicious intent

Common Attack Methods

Attack	Description
Malware	Malicious software (viruses, worms, Trojans, spyware, ransomware)
DDoS	Flooding a server with traffic to make it unavailable
SQL Injection	Injecting malicious code into databases
Cross-Site Scripting	Injecting code into trusted websites
Man-in-the-Middle	Intercepting communications between two parties
Password Attacks	Dictionary attacks, password sniffing, cracking programs
Drive-by Attack	Malware installed simply by visiting an infected website

10. Phishing

Definition

Phishing is a fraudulent attempt to obtain sensitive information (passwords, credit card details, personal data) by disguising as a trustworthy entity in electronic communication.

How It Works

1. Attacker sends messages that appear to come from a legitimate source (bank, employer, service provider)
2. Message creates urgency — "Your account will be closed," "Verify immediately"
3. Victim clicks a malicious link or downloads an attachment
4. Victim enters credentials on a fake website or installs malware
5. Attacker gains access to accounts or steals identity

Types

- **Email Phishing:** Mass emails to many recipients
- **Spear Phishing:** Targeted at specific individuals
- **Vishing:** Voice phishing via phone calls
- **Smishing:** SMS phishing via text messages

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11. Cyberbullying

Definition

Cyberbullying is the use of digital communication tools to harass, threaten, embarrass, or target another person. It can occur on social media, messaging apps, gaming platforms, or any online space.

Common Forms

Form	Description
Harassment	Repeated unwanted messages or threats
Trolling	Deliberately provocative or abusive comments
Impersonation	Pretending to be someone else online
Outing	Sharing someone's private information publicly
Exclusion	Deliberately excluding someone from online groups
Cyberstalking	Persistent unwanted contact or monitoring

Form	Description
Revenge Porn	Sharing intimate images without consent

Prevention and Response

- **Do not respond** to bullies — it often encourages them
- **Save evidence** (screenshots, messages) for reporting
- **Block and report** the bully on the platform
- **Tell a trusted adult** (parent, teacher, counselor)
- **Contact law enforcement** if threats involve violence or exploitation

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12. Overview of the Indian IT Act

Information Technology Act, 2000

India's primary legislation governing cyber activities and digital commerce. It provides legal recognition for electronic records and digital signatures and addresses cybercrimes.

Key Provisions

Section	Offence/Provision	Description
Section 43	Penalty for damage to computer	Unauthorized access, downloading, copying data
Section 66	Computer-related offences	Fraudulent or dishonest acts (read with Sec 43)
Section 66C	Identity theft	Fraudulent use of another's electronic signature, password
Section 66D	Cheating by personation	Impersonation using computer resources
Section 66E	Violation of privacy	Capturing/publishing private images without consent
Section 67	Obscene material	Publishing or transmitting obscene content
Section 67A	Sexually explicit material	Publishing material containing sexual acts

Section	Offence/Provision	Description
Section 67B	Child sexual abuse material	Publishing/transmitting CSAM
Section 69A	Blocking	Government power to block public access to information
Section 78 & 80	Investigation	Empowers police to investigate offences

IT (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021

These rules impose **due diligence obligations** on intermediaries (social media platforms, ISPs, websites):

- Prohibit hosting/transmitting unlawful content (obscene, invasive of privacy, harmful to children, promoting hate)
- Require **removal of unlawful content** within 24-72 hours of complaint
- Mandate **grievance officers** for user complaints
- **24-hour takedown** for non-consensual intimate imagery
- **2026 Amendments:** Require measures to prevent AI-generated unlawful content, labeling of synthetic media, removal within **3 hours** for sensitive content

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13. E-Waste: Hazards and Management

What Is E-Waste?

Electronic waste (e-waste) refers to discarded electronic devices — computers, mobile phones, televisions, refrigerators, and their components. It is one of the fastest-growing waste streams globally.

Why E-Waste Is Hazardous

Electronic devices contain **toxic substances** that can leach into soil and water:

Substance	Source	Health Hazard
Lead	CRT monitors, solder	Neurological damage, especially in children
Mercury	LCD backlights, batteries	Brain and kidney damage
Cadmium	Batteries, circuit boards	Kidney damage, bone disease
Hexavalent Chromium	Metal plating	Cancer, respiratory damage
Brominated Flame Retardants	Plastic casings, cables	Endocrine disruption

Environmental Impact

- Soil contamination — toxic metals leach into groundwater
- Air pollution — improper burning releases dioxins and furans
- Water pollution — heavy metals contaminate rivers and lakes
- Bioaccumulation — toxins enter food chains

E-Waste Management

Formal Recycling: Properly licensed facilities that safely dismantle and recover materials.

Extended Producer Responsibility (EPR): Manufacturers are responsible for the entire lifecycle of their products, including take-back programs.

Best Practices for Individuals:

- Donate working electronics to schools, NGOs, or charities
- Use certified e-waste recyclers (check government lists)
- Never burn or throw e-waste in regular trash
- Buy devices with longer lifespans and repairability
- Sell or exchange old devices through authorized channels

14. Awareness About Health Concerns Related to Technology Usage

Physical Health Concerns

Concern	Cause	Prevention
Computer Vision Syndrome	Prolonged screen exposure	20-20-20 rule: every 20 minutes, look 20 feet away for 20 seconds
Repetitive Strain Injury (RSI)	Repeated keyboard/mouse movements	Ergonomic keyboards, proper posture, regular breaks
Text Neck	Looking down at phones for extended periods	Hold devices at eye level
Sleep Disruption	Blue light exposure suppressing melatonin	Avoid screens 1-2 hours before bed; use night mode
Obesity/Sedentary Lifestyle	Prolonged sitting	Stand up regularly, exercise daily
Digital Eye Strain	Glare, poor lighting, uncorrected vision	Adjust screen brightness, use anti-glare screens

Mental Health Concerns

Digital Addiction: Compulsive use of internet, social media, or gaming that interferes with daily life.

Social Isolation: Preferring online interaction over face-to-face relationships.

Anxiety and Depression: Linked to social media comparison, cyberbullying, and constant connectivity.

FOMO (Fear of Missing Out): Anxiety about missing rewarding experiences others seem to be having online.

Attention Fragmentation: Constant notifications and multitasking reducing ability to focus deeply.

Healthy Technology Habits

1. **Set screen time limits** — use built-in tools to monitor and limit usage

2. **Create tech-free zones** — no devices during meals or in bedrooms
3. **Take regular breaks** — use the 20-20-20 rule for eye health
4. **Exercise regularly** — physical activity counteracts sedentary screen time
5. **Prioritize real-world connections** — schedule face-to-face interactions
6. **Practice digital detox** — periodic breaks from technology
7. **Seek help** if technology use is causing significant distress or impairment

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Summary Table: Key Terms at a Glance

Term	One-Line Definition
Digital Footprint	Trail of data left by your online activities
Netiquette	Rules of polite online behavior
Data Protection	Safeguarding personal data from misuse
IPR	Legal rights over creations of the mind
Plagiarism	Using others' work without attribution
Licensing	Permission to use copyrighted work
FOSS	Software with freedoms to use, study, modify, distribute
Cybercrime	Criminal activity involving computers or networks
Hacking	Unauthorized access to computer systems
Phishing	Fraudulent attempt to steal sensitive information
Cyberbullying	Harassment using digital communication tools
IT Act 2000	India's primary cyber law framework
E-Waste	Discarded electronic devices containing toxic materials

📘 **Exam Tip:** For Class 12 IP, focus on **definitions, distinctions (FOSS vs. proprietary, plagiarism vs. copyright), and the Indian legal framework (IT Act sections, DPDPA)**. Remember specific section numbers (66C, 66D, 66E, 67) and the four freedoms of FOSS — these are frequently tested.